

EVENT REGISTRATION & VENUE SCHEDULING SYSTEM

Design and Implementation of a
Database Management System

SUBJECT	DBMS
REVIEW	1
DATE	10 September 2026
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PROBLEM STATEMENT

Managing college events with manual or separate systems leads to inefficiencies across the event lifecycle.

Duplicate registrations

Event overbooking

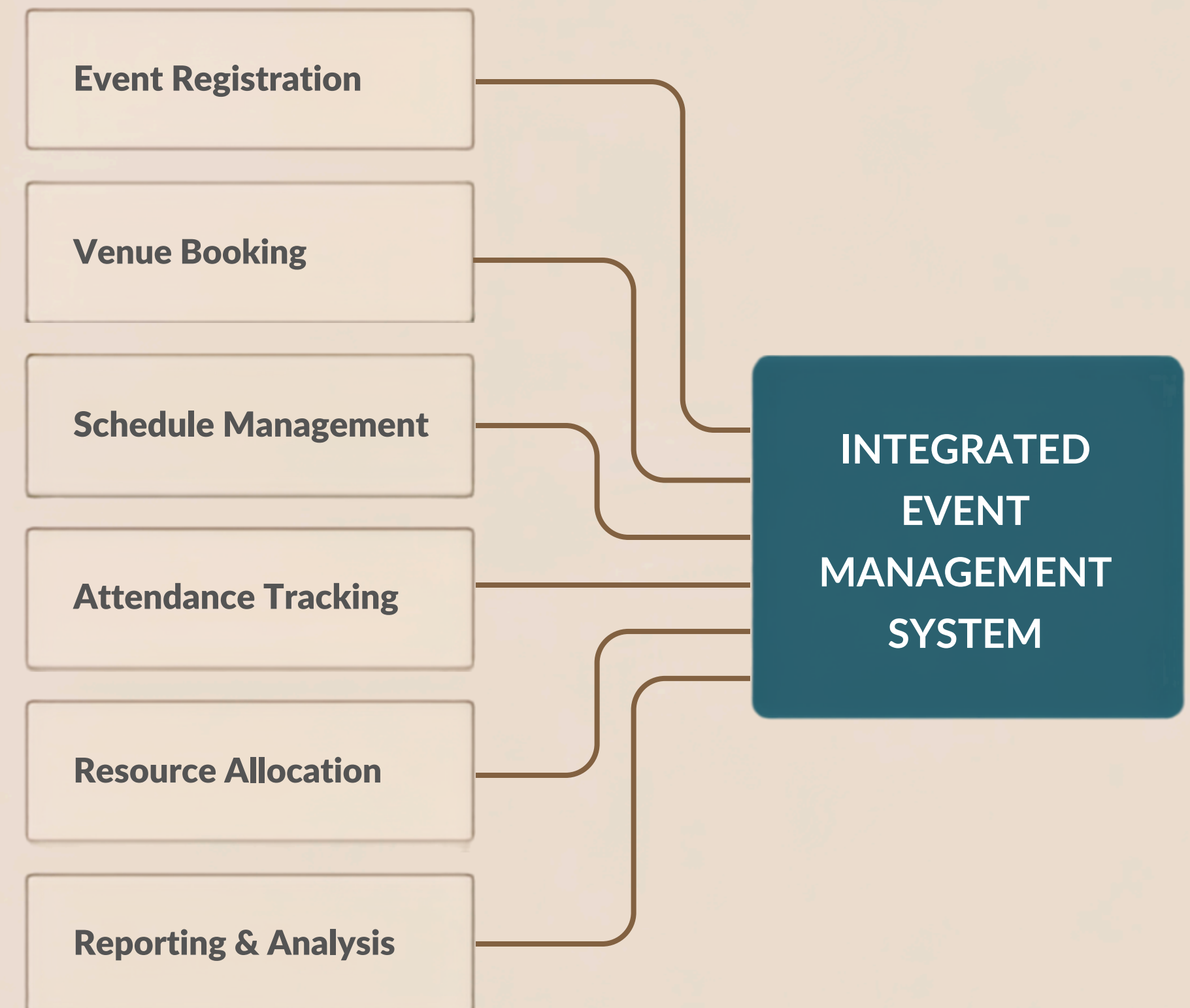
Venue conflicts

Poor venue utilization

Difficult attendance tracking

Difficulty generating consolidated reports

An integrated database system can help manage the complete event process efficiently.



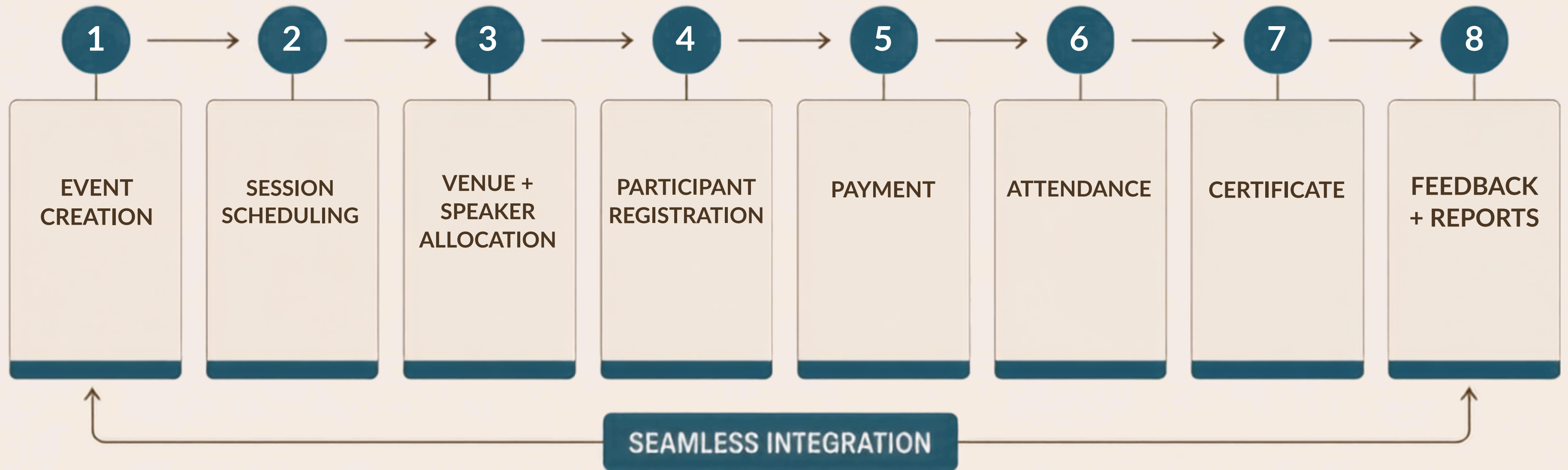
EXISTING SYSTEMS

Existing System	Main Use	Observation
Eventbrite	Event registration & ticketing	General-purpose platform
Cvent	Event & venue management	Commercial platform
Google Forms + Sheets	Registration & data collection	Requires manual management
College Event Portals	Event registration	Features vary by institution

Existing solutions provide useful event-management features, but our project focuses on an integrated relational database designed specifically for institutional events.

PROPOSED SOLUTION

A centralized relational database system for managing institutional events from planning to reporting.



OUR UNIQUE FEATURES



DYNAMIC WAITLIST & SEAT ALLOCATION

Additional participants
waitlisted when event is full

Next eligible participant
allocated seat when available

Automated capacity management



CAPACITY-AWARE VENUE ALLOCATION

Venue choice considers expected
attendance, capacity, availability

Prevents unsuitable venue
allocation and scheduling conflicts

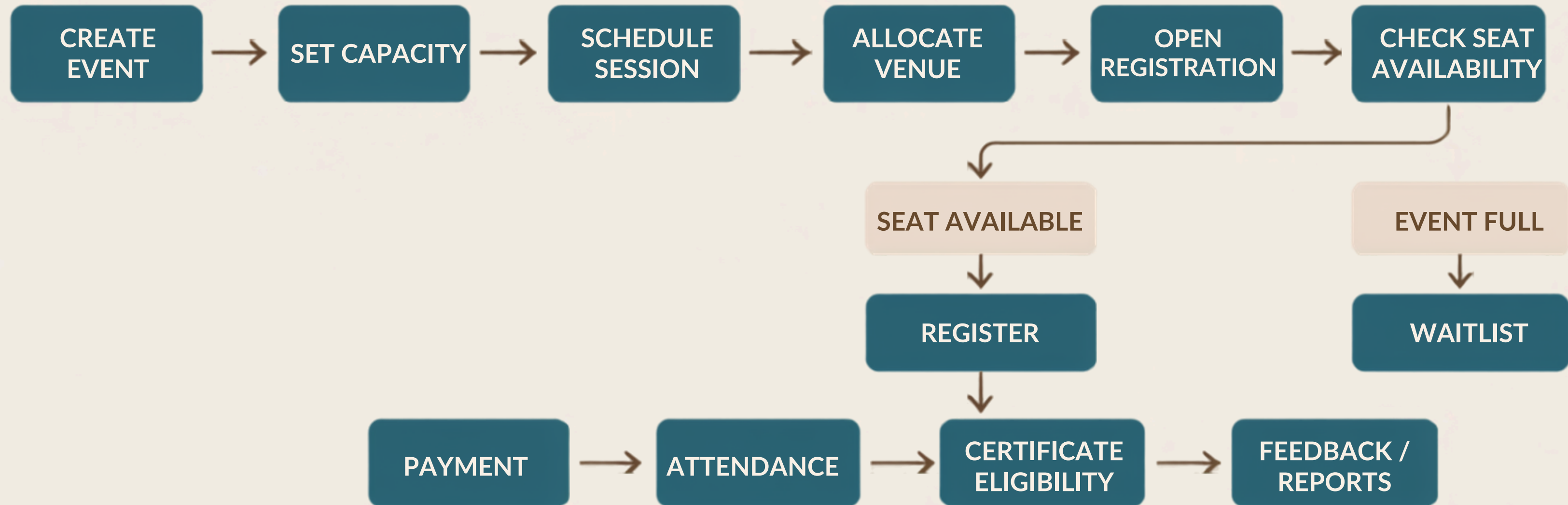
Conflict-aware scheduling

SYSTEM REQUIREMENTS

Core capabilities required for the Event Registration and Venue Scheduling System.

EVENT & SESSION MANAGEMENT	REGISTRATION	VENUE MANAGEMENT	ATTENDANCE & CERTIFICATES	PAYMENT & FEEDBACK
Create and manage events	Register participants	Check venue availability	Record session attendance	Record payment details and status
Schedule sessions	Prevent duplicate registration	Match venue capacity with expected attendance	Calculate attendance percentage	Collect participant feedback
Assign categories and coordinators	Check event capacity	Prevent overlapping bookings	Determine certificate eligibility	
Assign speakers and venues	Manage waitlist			
NON-FUNCTIONAL REQUIREMENTS				
Data integrity		Reliability	Security	Usability
Efficient data retrieval				

SYSTEM WORKFLOW & BUSINESS RULES



KEY BUSINESS RULES

1. No duplicate registration allowed

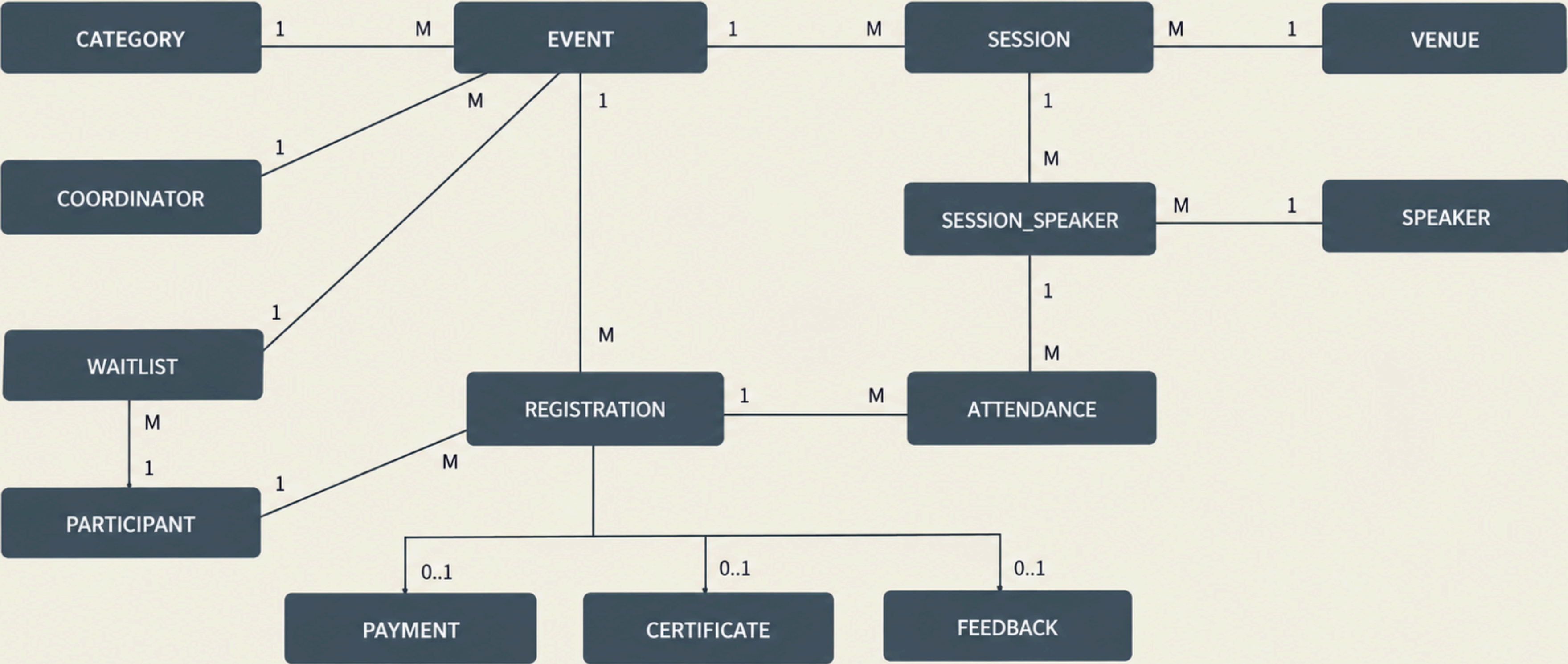
2. Confirmed registrations capped at event capacity

3. Venues cannot be double-booked during overlapping sessions

4. Venue capacity must accommodate expected attendance

ENTITY-RELATIONSHIP DIAGRAM

Overview of the main entities and their relationships in the Event Management System.



RELATIONAL DATABASE SCHEMA

Main tables with key attributes and relationships.

